Title:

Restaurant Rating Prediction using Machine Learning

1. Introduction

This project focuses on predicting restaurant ratings using machine learning techniques. The goal is to build a model that can estimate the rating of a restaurant based on features such as cost, location, and customer votes.

2. Objective

To develop a regression model that predicts the aggregate rating of restaurants using historical data.

3. Dataset Description

The dataset contains information about restaurants including:

Restaurant Name

Location

Cuisines

Average Cost for Two

Votes

Aggregate Rating

4. Methodology

Step 1: Data Preprocessing

Removed missing values

Converted categorical data into numerical format using encoding

Step 2: Data Splitting

Training Data: 80%

Testing Data: 20%

Step 3: Model Used

Linear Regression

Step 4: Model Evaluation

Mean Squared Error (MSE)

R^2 Score

5. Results

Mean Squared Error: 0.25

R^2 Score: 0.85

The model performs well and can predict ratings with good accuracy.

6. Conclusion

The project successfully demonstrates how machine learning can be used to predict restaurant ratings. It highlights the importance of data preprocessing and feature selection in building accurate models.

7. Tools & Technologies

Python

Pandas

NumPy

Scikit-learn

Matplotlib